

VITRUM: A WAY TO A TRANSPARENT SOCIETY

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Resum: L'objectiu de "Vitrum: a way to a transparent society" és la creació d'una base de dades innovadora i transparent que permeti a qualsevol usuari trobar informació objectiva i fiable sobre tòpics de política i societat, ja siguin actuals o passats, sense haver de filtrar les opinions parcials i evitant la censura que es troba present en els mitjans de comunicació tradicionals. Aquest sistema també permetrà el rastreig de transaccions amb diners públics mitjançant una stablecoin com a forma de lluitar contra la corrupció. Tot això serà possible gràcies a la pionera tecnologia blockchain, regulada per una institució que verificarà la fiabilitat i el correcte funcionament de la xarxa, precedit per una reforma legal per modernitzar els mecanismes reguladors de l'economia i el blockchain de l'administració pública en l'actualitat.

Paraules clau: Blockchain, Política, Red descentralitzada, Seguretat, Emmagatzematge en blocs, Base de dades, Transparència, Stablecoin, Fake news, Censura, Economia circular, Sostenibilitat, Sostenibilitat ambiental, Sostenibilitat econòmica, Sostenibilitat social, Vitrum, Digital twin

Abstract: The goal of "Vitrum: a way to a transparent society" is to create an innovative transparent database for anyone to find objective, reliable information about any political and social topics, current or past, without having to filter amongst biased opinions or political censorship that may be present in traditional media. This system will also allow the tracking of public fund transactions through a stablecoin as a way to fight against corruption. This will all be achieved thanks to pioneer blockchain technology regulated by an institution which will verify the reliability and well functioning of the whole network, following a legal reform to modernize the public administration mechanisms that regulate the national economy and blockchain technology as of today.

Keywords: Blockchain, Politics, Decentralised Network, Secure, Block Storage, Database, Transparency, Stablecoin, Fake news, Censorship, Circular economy, Sustainability, Environmental sustainability, Economic sustainability, Social sustainability, Vitrum, Digital twin



1.1 Main report content

1 INTRODUCTION

BLOCKCHAIN TECHNOLOGY is an innovative way of storing information in a database using a distributed and decentralised format. This system allows for a secure way to record transactions where upon completion, there is no way to modify the transaction in question. We have identified this innovative technology as a crucial tool to make a transition into a more transparent society.

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- **State of the Art:** An in-depth analysis on the context of the topics which will be treated in the paper. You can find information about the current state of journalism and fake news, from traditional media to modern "amateur" journalism. In this section there is also information relevant to the blockchain technology, its origins and legislation in Spain, stablecoins and circular economy.
- **Methods and Tools:** An overview of the methods and tools used to create this paper.
- **Proposal:** A first explanation of our project itself where the main goals are presented.
- **Application:** The practical application of our proposal in terms of the blockchain network and its regulation.
- **Discussion:** An analysis of our project from different points of view such as citizen science or the quadruple helix.

2 STATE OF THE ART

2.1 Journalism Nowadays

Since 1948, when Shannon and Weaver published the book "A mathematical theory of communication": it has been understood that communicative models are one-directional, meaning the subject is a passive member of the process of communication. While this theory is and has been crucial in the field of communication systems, when we extrapolate it to journalism and media communications, it has become obsolete [12].

Traditional media like newspapers and television have had the monopoly of information for decades, and that has led them to adopt a position of power [32]. At the end of the day the professional journalist was the only one qualified to spread news until a few years ago, so in order to gain more reputation they had to either get the best, most interesting, most recent and most exclusive news, or adulterate them to make them look like they met these requisites.

But in recent times the scene of media and journalism has changed a lot with the rise of social media as a new source of information. The biggest change that came with this transition from traditional media to digital media was that the profession of journalist became a bit blurry. Nowadays anyone can hop into any social media with an article they made up in 10 minutes and disguise it as if it was a real breaking news article. This rise of social media also means that now, in many cases, the communication is bidirectional, because the media can receive feedback through comments in various websites or social media.

2.1.1 The professional journalist

With the rising popularity of digital media, many sources have been taking advantage of the relatively easy money the internet offers, with many companies that work as a normal newspaper but who post all their articles in their websites or social media. This offers a wide variety of options when one goes into an internet browser to look for a specific type of news, allowing readers to find articles about anything they want.

This point is crucial because the user will be more or less willing to read, listen and comment on a news article depending on their personal need and not depending on the primary target of the news [12]. Therefore, for an article or document to satisfy a need for information on top of really informing and being relevant, it must have the desire of the final consumer for it to be valuable [12]. This is the reason why techniques such as click-baiting have risen in popularity over the years, the content doesn't really matter as long as you have enough clicks in your post.

The problem comes when digital media companies start abusing this system, and instead of providing more serious, contrasted, perhaps objective information, they choose to appeal to the reader's emotions and bait them into clicking their article before they can read that of a competing news source. When we factor in the immense number of news sources making use of these practices, we get users who are mostly incapable of contextualizing all news due to the overflow of inputs of segmented information which make them more and more vulnerable to fake news, by not being able to contrast any information in real time [12].

At the end of the day, this means the professional journalist has to work in the middle of this reality, therefore having two options: either join the "amateur journalists" or gain a reputation of offering accurate and reliable information to their readers.

2.2 Fake News

The problem of "fake news" has recently emerged, and it has quickly become a headache for some news consumers. The term fake news, which has become a widespread buzzword used by politicians and journalists alike, refers to news that can be verified as false and are intentionally generated to mislead readers [26]. In the news industry, as well as in social media, fake news detection and identification of news sources has become a central topic of discussion.

2.2.1 The power of misinformation

At first glance, it may only seem that fake news are unoffensive or that they can be easily identifiable, but that's far from the truth. In fact, they are considered to be a strong weapon in mass manipulation, which can be used to destabilize a country or polarize the public opinion. The power of misinformation is so big that it has been used in some of the biggest international political events of the last decade.

In June 2016, the result of the referendum on UK membership of the European Union (EU) was apparently slightly in favour of Brexit. This happened despite a the support of most experts to remain in the EU and tons of evidence that led to thinking that remaining in the EU was the right choice. The referendum process was filled with questionable, manipulated or biased facts on both sides, even though it was particularly clear in the Leave campaign, and by accusations of outside interference in the democratic process by external actors who were unaccountable for their manipulative actions, such as the Russian government [15].

As Rogers said in 2020 [28], "Much work has been undertaken on the influence of Russian (and Russian-style) influence campaigning in the US, and the presence or absence thereof during elections in Germany, France, Sweden and elsewhere". Or as Reston said in 2017 [27], "For years, the Russian propaganda machine — a loose network of hackers and state media outlets, Twitter bots and bloggers — has pumped out a steady stream of digital disinformation aimed at drumming up support at home and destabilizing enemies abroad".

So as we can see, and as is supported by evidence and experts, fake news and media manipulation are already being used as an extremely powerful weapon by one of the largest most influential powers in the world, so much so that many governments have used it as well in favor of their interests.

2.2.2 Media censorship and press freedom

Moreover, on top of the exaggerated amount of fake news out there, there's also the problem of press freedom and censorship. Each year, a world press freedom index is released by reporters without borders, taking into account

plenty of fields such as transparency, media independence, or pluralism of opinions in the media. In their 2021 report, it was determined that only 48 of the 180 included countries were in a good or at least satisfactory situation, while 58 of them were in a problematic situation and the remaining 73 were in a difficult or very serious situation. It doesn't make things any better that some of the world's most influential powers find themselves near the bottom of the rankings, with many important European countries and up and coming developing countries also ranking fairly low. Countries such as Hungary or Turkey are notable examples because they have passed censorship laws recently and got into a little trouble with their neighboring powers, but ultimately carried on with those laws unpunished. Then there are way more extreme examples like China, India or Russia being powerful nations with extremely low press freedom, and the middle east in general being an abyss of very serious censorship. The global situation of press freedom is represented in figure 1.

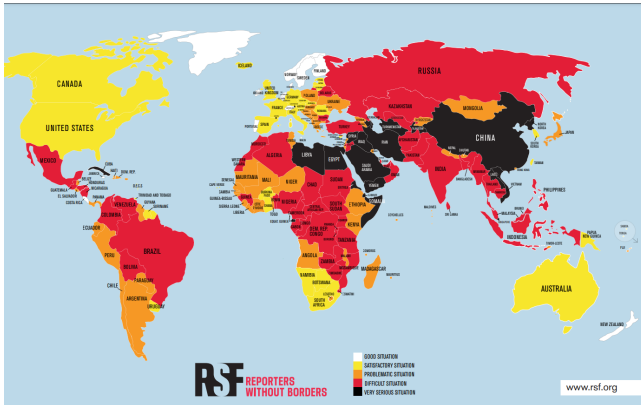


Fig. 1: RSF Press Freedom Index map, 2022

Source: Reporteros sin fronteras España ¹

This goes to show that censorship and fake news are closely related, because in all these countries where the situation is problematic or worse, even in the countries near the top, the information published by the national media is clearly biased towards a certain political ideology or institution, meaning the people will have a hard time finding reliable information, or in some of the most extreme cases of censorship they won't be able to find anything other than what the government wants to show them.

In conclusion, we have seen fake news are a dangerous weapon that can easily manipulate masses and help political parties or governments achieve their interests which under normal circumstances people would be against, but with the help of censoring the opposition's voice and sending out misinformation the general public stays in favor of those interests.

2.3 Blockchain

Blockchain is the term that defines a technology that was born along with Bitcoin. It is based on P2P (Peer to peer) technology, a network of computers that work for a main

purpose and each node helps in the overall performance of the network. These nodes can be distributed in different areas and do not need to be in the same room. The users in this network can transfer information without the need of any intermediary between, basically: each computer is connected to other computers. This technology grants different strong points:

- **Decentralization:** each node is equal to the other so each one is important for the good network performance.
- **Robustity:** Allows the exchange of information without any intermediary between.
- **Scalability:** As more nodes are connected to a P2P network, the better will be the network performance, because each one shares their own resources. This is different from the architecture client-server, in which with an increase of petitions can affect the experience of the user.
- **Distribution of costs for every user:** Some resources are exchanged for some others, like files, bandwidth or disk space for example [31].

So Bitcoin uses this technology behind. A chain of blocks registers all the interactions of the network and its users. It uses hash functions that ensure the privacy and anonymity of the network. The proof of work (workforce of the network) ensures that new coins of that network are created and the miners (computer farms) are rewarded for participating. So every user can be rewarded for participating in this decentralized finance network without any restriction or limit, granting decentralized public but anonymous transactions without intermediaries. In the counterpart, the banks instead, only work 8 hours a day and on the weekend they close their offices, all the transactions are verified by them and your account can be blocked.

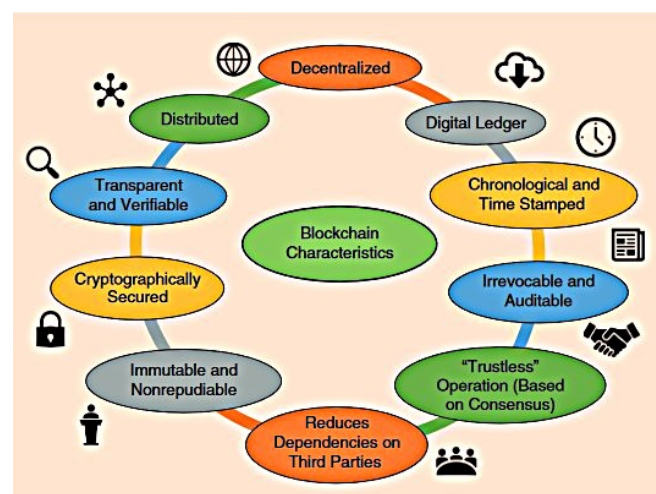


Fig. 2: The pivotal characteristics of a blockchain [25].

2.3.1 Origins of Blockchain

Satoshi Nakamoto, the creator of Blockchain, thought of this network because of the following aspects:

¹Reporteros sin fronteras: <https://www.rsf-es.org/clasificacion-2022-tabla-de-paises/>

- Shortcoming of the monetary system
- Lack of privacy
- Centralization boom

The true origins of the Blockchain are the P2P networks, the first of which became operative in 1999. Napster was an innovative project developed by the American entrepreneurs Shawn Fanning and Sean Parker, who popularized the concept of a distributed network between users, introducing an application to share music and files. It achieved 26 million users while the internet had 248 million users. Later on, this company was sued because of copyright infractions and the platform was closed. Napster was the start of the P2P networks that we know nowadays [30].

At the beginning, a single Bitcoin had a sale price of 0,03 USD, and during recent years, it has achieved a value over 50.000 USD, making the Blockchain technology even more appealing and popular for some technology sectors. This is why some businesses have been adding this technology in their digital layer recently despite it already existing 12 years before Bitcoin was created [17].

2.3.2 Different types of Blockchains

When a blockchain is public it means that anyone can participate and read the transaction book. They are open to anyone and everyone can participate, make it better, develop or check them. These blockchains are completely decentralized and there is no authority that rules their performance. There is a relation of equality between nodes. The maintenance of the blockchain depends on the mining activity and commissions of transactions. The performance of the network is transparent and everyone can participate. A Blockchain is private (permissioned blockchain) when it is managed by an authority and the possibility to participate is limited to a list of participants. The authority is responsible for the blockchain maintenance. The access to the book of transactions is private. An hybrid Blockchain is when a public blockchain encounters a public and a private blockchain with the aim to benefit from these two sides of a coin. Everyone can check what is in the blockchain but participation is restricted. This is a very useful choice when governments or institutions want to store their data in a safe way. It is partially decentralized to ensure security and transparency. The access is restricted [24].

2.3.3 Security and Privacy on Blockchain

As we have said before, Blockchain is a decentralized technology that allows transactions. This technology offers the possibility to be securely and verified without the necessity of any centralized technology. Security and privacy are ones of the main keys of the blockchain technologies and have the power to cause the success or failure of this technology [13]. We can see the process of a transaction in the following scheme:

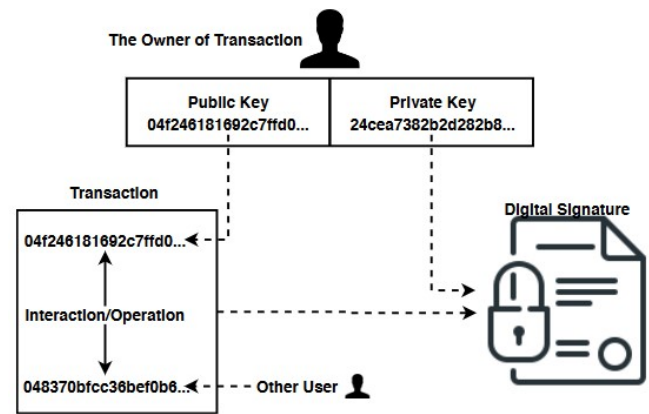


Fig. 3: Simplified scheme of a blockchain transaction [5].

2.3.4 Private Key encryption and transparency

In order to provide security when we browse the internet or use machines and networks, it is very important to focus on Asymmetric keys. In blockchain technology the keys play an important paper to identify the participants of the network and controlling the assets in the blockchain network. Each network participant has assigned a key pair. However the most important things as transfers are managed by using the private key. Digital wallets are very linked to asymmetric keys. The digital wallets is where each participant of the blockchain network has their cryptocurrencies stored. Digital wallets are the responsible of the creation of public and private keys pair. The private key is encrypted and stored on the servers [5]. By taking into account these data, we can affirm that blockchain technology provides transparency and anonymity. Every person has their own private key and the digital wallet is described as a long sequence of digits and letters that does not give information about the name of the people who are involved in the transactions. only the person who sends and who receive know about the other.

2.3.5 Cybersecurity

When we talk about cybersecurity we are referring to the measures for the preservation of confidentiality, integrity and availability of information in cyberspace (by ISO27032/2012) ². In the blockchain technology it is considered that confidentiality of users of the network is high due to cryptography of public key that code the transactions. The different blocks are linked between them and contain information of other blocks. The algorithm is very difficult to hack and very expensive material is needed and ends up being more expensive than the eventual profit, the way to hack the system is to control the 51% of the blockchain network. If we remember that blockchain is a decentralized technology it is easy to see that the attack should be huge. Imagine how difficult it could be to hack millions of computers around the world. It is widely considered that blockchain technology can enhance the security of many transactions. However one possibility

²ISO/IEC 27032:2012 provides guidance for improving the state of Cybersecurity, drawing out the unique aspects of that activity and its dependencies on other security domains

for the hackers or criminals would be to hack internet and thus block the network and not let the different users interact. Some recommendations that anyone has to take into account in their interaction with digital environments can be: Using robust passwords, with possible phrases mixed with symbols that do not have to do with aspects of personal life, implement 2-step verification in the platforms, profiles and programs that are use, use encrypted hard drives and pen drives, Use updated operating systems with antivirus and firewalls, Make backup copies frequently on different media, don't not store crypto in hot wallets or exchanges, if possible use a cold storage. The concept of hot wallet is used to refer to those cryptocurrency wallets or virtual wallets that store private keys in places that are always connected to the Internet. The opposite of hot wallet is cold wallet or cold storage [6].

2.3.6 Smart Contracts

A contract is an agreement between two parts written in a piece of paper. A set of rules are defined and they have to be accomplished. A smart contract is a contract executed by a computer with a set of rules. There are specific rules that everyone accepts so they can understand what kind of interactions will work between them. The innovation on this invention is that the computer does not need to trust the other node to interact with them, they just need to run the smart contract, so you do not need to be asking if one node is trustable or not, in fact they have to run the same piece of code to be trustworthy. In more detail, a smart contract is a piece of code in which the rules of interactions of a blockchain are defined and the punishments in case of a bad use of the blockchain, and they are stored on every node. The use of Smart contracts in a blockchain provide some benefits like: Trust, Security, Autonomy, Speed and Precision. But the Smart Contract has to be precisely designed, because an intruder can exploit a breach of the system and attack the whole digital system. Leading to a loss of money and trust of the network, and this could be the end of a blockchain, most of the attacks in internet are because of a failure or a vulnerability on the system [24].

2.3.7 Decentralization

We can say that the blockchain technology is decentralized because it does not have a specific location and each computer or machine in the entire network is interrelated with each other and consolidates the entire system [25]. On the other hand, we can divide the decentralization in blockchain technologies in three different parts: decentralized ledger, decentralized smart contract and decentralized applications. First of all, if we talk about decentralized ledger, it is mandatory to discuss the topic of Bitcoin. Bitcoin is one of the most important cryptocurrencies that has benefited the blockchain to expand and become more indispensable. Secondly, smart contracts are becoming more and more common by using the platform of Ethereum, a special platform designed to facilitate decentralised smart contracts. The idea of smart contract is basically that legal contracts can be do it automatically reducing the time and easily. Furthermore, the decentralization of blockchain technology offers an

amount of different applications very useful and practical. For example in terms of Finance Services, Smart properties, Internet of Things (IoT), Smart Health Care or Smart Government [8]. The difference between a centralized system and a decentralized blockchain system is represented in the figure 4. Furthermore, the different uses of a blockchain technology are represented in the Figure 5.

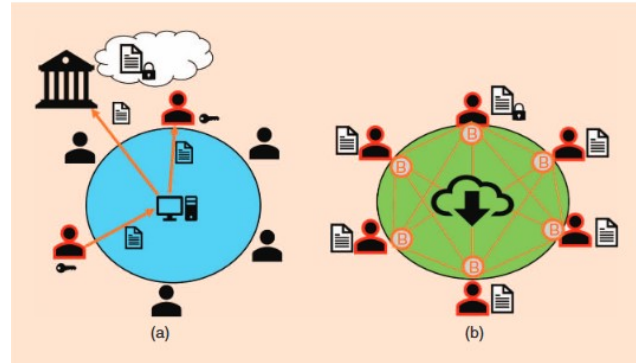


Fig. 4: The (a) centralized system with intermediaries versus a (b) decentralized block-chain system [25].

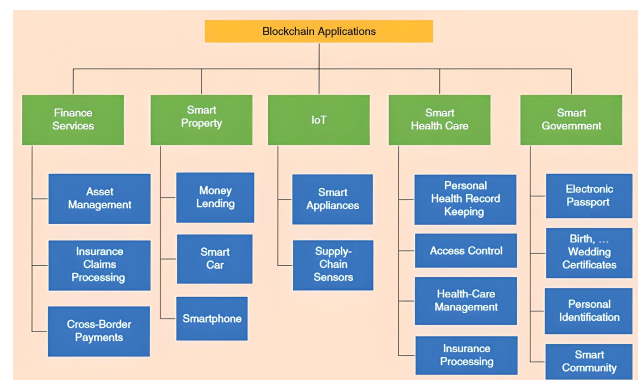


Fig. 5: The potential applications of a blockchain [25].

2.3.8 How can blockchain technologies make the political system more transparent?

Due to the lack of transparency of our current system, it is not strange to see more and more cases and investigations in the topic of corruption. There are reasons to believe that if Blockchain were to be used as the main system to regulate the economy, corruption would decrease and the public administration would be more reliable because it would be more transparent. Without using Blockchain, corruption cases can be hidden relatively easily because record keeping is done by central authorities, but blockchain provides a unique combination of permanent and tamper-evident record-keeping transaction transparency, auditability and reduction of the centralized authority, making it easy to trace public funds and harder to disguise corruption [1]. When companies and government have a non-transparent ground they can corrupt, and this affects the political system because corruption undermines the legitimacy of a government and democratic values such as

trust and tolerance [9]. If blockchain brings transparency into government processes, that gives the citizens better decision-making chances, and the political system can be more reliable. For that purpose, Inter-American Development Bank (IDB) has a partnership with the Office of the Inspector General of Columbia to design and use blockchain technology to surveil corruption-prone government processes. This project is called Unlocking Government Transparency with Blockchain Technology [18].

2.3.9 Mining cryptocurrencies

The cryptocurrency system is getting more sustainable every day. Blockchain and cryptocurrencies can help the environment. The traditional financial system requires significantly more energy than Bitcoin Network. Still Bitcoin Network requires energy as well, but it is a new network, and it is developing and becoming more sustainable and greener. For example, when the Ethereum 2.0 is created the model shifted from proof of work (PoW) to proof of stake (PoS) model. PoW is a decentralized system that provides the power for the Bitcoin Network, but with that model high amount of energy has been used. PoS requires significantly lower amount of energy than PoW. Some calculations and estimations show that PoS model will be 99% more energy sufficient than PoW [33]. There are some other solutions for making the cryptocurrencies more sustainable, like proof of capacity, proof of history, proof of elapsed time. These are more efficient than current PoW model. Proof of capacity model employs spare space on device's hard drive to store solutions to a cryptocurrency hashing problem and it is 30% more efficient than current model. With proof of elapsed time model it is a bit like PoW, but still, it is more efficient than that because with proof of elapsed time model the miner's processor goes to sleep. Cryptocurrencies were adding approximately 40 million tons of carbon dioxide to the atmosphere and that's why it has a huge carbon footprint, but as an example Bitcoin are rapidly trying to use renewable energy sources and now the 56% of its energy usage is provided by renewable energy sources. There are some laws and taxes in the favor of the environment like federal solar tax credit. It allows 30% of deduction from the installation cost of the solar energy system [14]. Another improvement is having new kind of a token that you can mine with the wasted energy.

2.3.10 The first Spanish blockchain technology

Alastria is the first Spanish institution to create a private or authorized blockchain in Spain. The purpose of these organisms is to allow all the registered companies in the Alastria system to write down commercial and industrial operations, guaranteeing both the identity of the operators and the file or registry by using cryptographic technology. All this information will be saved and stored in different blocks of the blockchain. Any company or institution can join this network but to interact with others and publish information Alastria have to authorize previously that the new user can register transactions. It is compulsory for the companies that want to be part of these infrastructure

to know some information about the platform and to abide by a set of rules: Firstly, the company has to be domiciled and constituted in Spain because this consortium has been thought to promote and strengthen the Spanish business and institutional fabric exclusively. Second, any type of company can be part of these consortiums, such as startups, SMEs ("pymes") or big companies (IBEX-35). Third, the platform is not for profit but the common interest in developing the infrastructure and improve the companies. Furthermore, there exist own rules within the platform about relations between entrepreneurs, customers, suppliers or stakeholders. In addition, being an open platform, competitiveness between companies is enhanced and usually smart contracts are common. Moreover there are different types of companies depending their own objectives. For example it is not compulsory to write and publish information, also it is possible to only read or validate or consult information about the others. To summarize, it is a good way for different companies across the country to collaborate and compete in order to achieve the majority of their goals. The main advantages of Alastria platform are: reducing documentation costs, increased speed of transactions, improvement of the conditions of competitiveness in the markets and legal security, digital identity and economic efficiency [20].

2.3.11 Blockchain towards smart cities

More than half of the global population lives in urban areas, and is predicted that in 2050 2.5 billion people will move to cities. The concept of "smart city" appears with the aim to response to all the problems that will affect cities in the following decades as traffic jams, pollution (waste and gas emissions), economics, politics... If we focus the topic of smart cities in a sharing economic perspective we can summarize it with the following graphic:

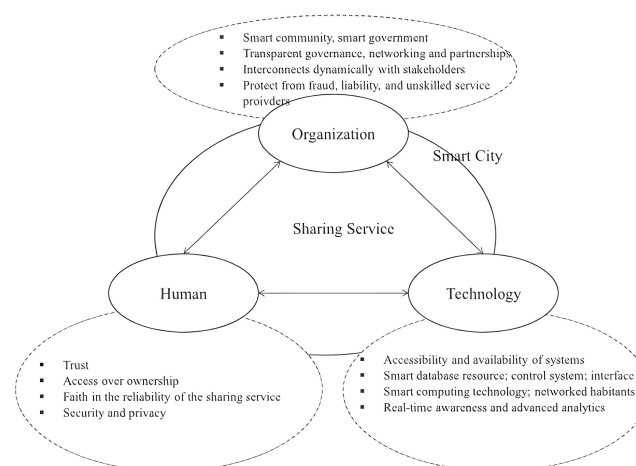


Fig. 6: A conceptual framework of smart city

Source: Blockchain-based sharing services: What blockchain technology can contribute to smart cities [34].

Smart cities are concerned about humans, technology and organization. The technology dimension is based on "information and communication technology" (ICT) to improve life and work of the city. Further, human dimension is about people education, learning, and knowledge. Finally, organization is about policy and

governance between governments and stakeholders. How we can achieve the objectives and the dimensions of the smart cities? Blockchain is a good way to manage all this points as we see in the figures 6 and 7. We can see outside

power consumption than credit card transactions and this kind of power requirements leads to a huge energy consumption which makes a bad effect on the environment [11].

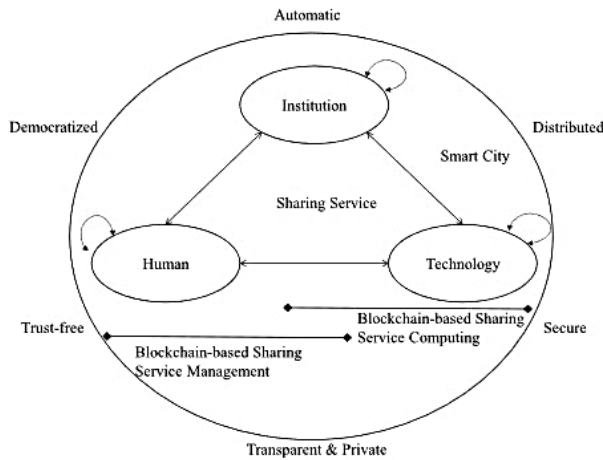


Fig. 7: Features of the management and computing of blockchain-based sharing services

Source: Blockchain-based sharing services: What blockchain technology can contribute to smart cities [34].

the circle the characteristics of blockchain technology: automatic, distributed, secure, transparent and private, trust-free and democratized that envelop the three dimensions of smart cities. In other words, with blockchain dimensions we can offer solutions to smart cities dimensions. In the field of smart cities With blockchain we eliminate intermediaries and increase the efficiency of a sharing service. Moreover, trust-free sharing services has been opened due to the opportunity that blockchain brings to new styles of digital interactions. Transparency and Privacy is the key to set up a relation between human and technology. In addition, democratization by blockchain technology can offer a good relation between humans and organizations because people can give their opinion and vote freely. The automatizing is also very important in the process of uniting blockchain and smart cities because it turns the system into a faster system with more capacity and possibility to deal with strangers while maintaining the privacy and anonymity of citizens. If we are able to develop and extend all this points not only theoretical but practical it will be the good way to convert cities to smart cities. Including blockchain technology open a lot of possibilities to improve the current cities and the entire world [34].

2.3.12 Some areas that can be improved of blockchain

Even the new block chain technology has already developed a lot it is still not perfect and needs to be improved more to be the main one. For example, the blockchain needs to be accessible to a wider public. Block chain and cryptocurrency need to remain decentralized and adopt for the whole world for taking the current system. Also, transparency of blockchain is a debatable topic because with that it is easier your competitive business know your business strategy. Bitcoin can make less transactions per minute than credit card transactions, this can cause transactions delay and Bitcoin transactions require more

2.3.13 Business implications of blockchain

Blockchain helps the current industrial systems evolution by providing an increase of privacy and security of the transactions and improvement on data collection, asset management, supply chain management and healthcare. This improvement on the current industrial system leads to a redefinition on businesses strategies and their activities. Businesses has been changing their activities from traditional business models into new ones [35]. Lately some startups throw around the term “Cryptocurrency Revenue Business Model” These means that the new generation is starting to use the new blockchain model instead of the traditional model and there are some crypto-based business models that already proved that they could have revenue with this new model. Many businesses try to adopt to these new blockchain systems because if they don’t in the future, they will lose their market activities. There are several crypto business models like: Token Development amp; Exchange (Ethereum), MLM or Altcoin Development, just to mention a few. Most of the startups are using one of these business models and lots of big companies also are changing their business model or at least they accept cryptocurrency such as Microsoft, Burger King and some parts of Amazon.

2.4 Royal Decree-law 14/2019, of October 31st

The main legislation which regulates blockchain in Spain nowadays is present in the Royal Decree-Law 14/2019, of October 31st, which adopts urgent measures for reasons of public security in the matter of digital administration, public sector contracting and telecommunications [10].

In the society where we currently live, there is a development of new technologies and communication networks that is accelerating to extraordinary levels, there are new trends in these branches every time; That is why we must establish a legal framework that allows us to guarantee public security, so that the digital administration is used for legitimate purposes that do not compromise the rights and freedoms of citizens, and that they are not negatively affected.

Technological development implies greater exposure to new threats, such as “cyberspace”, such as data and information theft. Today’s hyperconnectivity requires better protection of networks and systems, including the privacy and digital rights of citizens. Currently, this royal decree-law consists of different chapters where it contains laws to regularize everything. Still, some parts of these laws have been changed to improve safety. Some measures have been chosen:

- Measures regarding national identity documentation
- Measures regarding electronic identification before Public Administrations, to guarantee public safety.

- Measures to strengthen security in telecommunications.
- Measures to strengthen coordination in terms of security of networks and information systems.

They all have a lot in common, and this has been a very positive point for the Blockchain, since it allows us to see people's transactions, but without the need to know who that person is or the reason for said activity. Thus the security/intimacy of people is safe (without taking into account the accounts of politicians, who are in a separate case). On the other hand, this requires a much more advanced and secure security. The changes that are to be made in these laws comply with the principles of legal certainty since they are consistent with the rest of the national legal system and the European Union. After the adoption of these new regulations, improvements can be seen in all areas, from electronic administration, telecommunications and personal safety [10].

2.5 Stablecoin

The cryptocurrencies which are useful as method of payment are the elemental part of most blockchain technologies. It depends of the blockchain but for example in the case of Bitcoin, the token is a cryptocurrency and its use as a means of payment is its primary purpose. In other cases, the token serves as a "utility token" to perform transactions on the blockchain. Nowadays, the existing cryptocurrencies are too volatile to be a good medium to store or exchange values [29]. Stablecoin is a type of cryptocurrency that is less volatile than others. The idea of stablecoins are to simulate the original value of money or assets with the aim to minimize risks and convert the physical money to digital money without lose the value. The stability that offers stablecoin exchange make its investment attractiveness. Moreover, it allows us to consider it as a potential method of payment, saving and exchange of assets. There are many countries that are planning to create their own tokens so they want to use stablecoins. Great examples of them are Venezuela that has created a cryptocurrency called El Petro which its price corresponds to the price of a barrel of oil set by OPEC for the previous day of the end of trading. Another example is the Central Bank of the Russian Federation that is planning to create a cryptocurrency to control the reliability and continuity of payments [4]. It is also important to distinguish between two different concepts settlement of stablecoins:

- Retail stablecoins: assets available to a wide range of users.
- Wholesale stablecoins: assets available to a limited circle of users [4].

As we have said before, stablecoins are digital currencies non-volatile so are very transparent and every transfer is stored on a public blockchain. We can see the opposite of non-volatile in the following graphic:

In this way it is easy to track the flow of transaction and it could be used as a good security option to control and protect the entire system [2]. If we can establish guarantees for the stability that stablecoins offers, this would be the

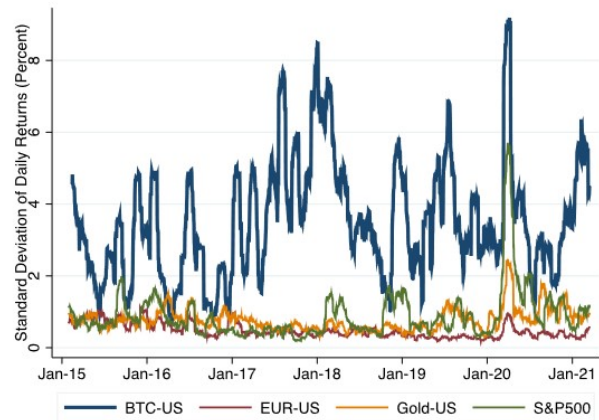


Fig. 8: Daily Volatility The 30-day (rolling) standard deviation of daily USD price changes of Bitcoin (BTC-US), Euro (EUR-US), SP500 stock market index (SP500), and Gold (Gold-US). All data is from FRED, Federal Reserve Bank of St. Louis. 2015-01-15 to 2021-03-16
Source: Currency stability using blockchain technology [29].

good way forming a strong decentralized financial system and facilitating economic adoption of cryptocurrencies [21]. One popular StableCoin is Tether USDT, which simulates the price of a dollar. It is a way to convert your fiat money into cryptos but without suffering from market volatility.

2.6 From physical to virtual payment

2.6.1 Reasons of the rise of digital money

We have to distinguish between the users and the producers. For the first group, digital money is very secure and comfortable. On the other hand, for the producers, the reasons reside in the low cost of operating with this money. Moreover, the use of digital money helps administrations to control the flow of the money and reduce the use of cash that stimulates bad practices as the payments by black money or dirty money. With the use of digital money it is easy to track routes of money flows and detect anomalies in the system. The Bank entities play an important role in this paper. Are the entities that control the most part of the market and its decisions can directly affect the transition from physical payment to virtual payment. It is known that in the coming decades physical money will tend to disappear and consequently virtual money will increase. [19].

2.6.2 Changes in payment systems: from a physical means of payment to an account currency

Until now, the physical means of payment with banknotes and coins have been the predominant in the world. Even so, today the monetary authorities seek greater efficiency for the future, since payments by card (non-physical) are currently faster and safer than making them with bills or coins, for this reason it is growing. This can be a point in favor of the Blockchain since the payments and transactions that they carry out are not physical [19].

2.6.3 Bitcoin vs credit card transaction

Throughout the time from the past the way of payment between people has been changed. Firstly, it was trading their valuable stuff, after that it turned to physical payments and until the 50's different types of physical payments are used. Then physical payment turned into card payments and nowadays card payments are turning into crypto currency payments. In the future it is expected of a decrease of the usage of credit card payments. There are some few reasons about this change. One of them is bitcoin transactions operate more like a cash because there is no financial intermediary, and the exchange is person to person. By credit card transactions there are also a lots of charge fees that banks get. The customers don't want to pay more money these charge fees. Also, Bitcoin transactions are much safer than credit cards because it is much harder to hack and with credit cards it can also be stolen [7].

2.6.4 Is Bitcoin/cryptocurrency ruining the economy?

Some people see cryptocurrencies are a threat for the economy and others think that the cryptocurrencies are a game changer. Cryptocurrencies are becoming the main factor of the economy. In the past that was cash, then it transitioned into credit cards and now it is evolving to cryptocurrencies. As with everything, cryptocurrencies also have some advantages and disadvantages. One of the main points in favor is being easily accessible. Almost everyone has their mobile phones and devices and with cryptocurrencies it is accessible through those devices and makes it easier to reach most of the people. Also, the transition cost of cryptocurrencies is very low or none because it is decentralized and that helps people save some money [22]. Another great advantage they have is transparency given by this easy access to information about transactions. On the other hand, cryptocurrencies have disadvantages as well. Some small and traditional companies are going bankrupt because of the economic change brought by the cryptocurrencies. Also, an economist called Eswar Prasad finds this change risky and complicated. In his opinion cryptocurrencies are going to make wealth inequality worse, as he quoted "Cryptocurrencies may contribute to monetary and financial instability, especially if they were to spawn a large and unregulated financial system that lacks investor protection" [16]. To sum it up, cryptocurrencies are not ruining neither helping the economy yet. They have advantages and disadvantages and the transition to this new model is harmful for some people and beneficial for others, but with the developing rate of this new technology it seems inevitable that sooner or later cryptocurrencies will be a staple in the market.

2.7 Blockchain and Circular Economy

The main goal of circular economy is minimizing the waste per used resources (basically making products with more sustainable materials and recycling the waste or giving it a second life) and developing the economy around this concept. Blockchain technology supports two of the main uses of the circular economy which are proving product origins and incentivizing positive behavioral change. The

transparency of the blockchain helps circular economy because it can provide trusted information about the things we buy and where we buy them from, and this puts a great pressure on companies to change their production methods for more sustainable ones. That transparency also provides positive consumption. Blockchain also can tokenize the natural resources and give them a digital value and identity and with that the value of the resources would be more apparent. Moreover, it can provide digital tokens for waste like plastics, so that they can be used again [23]. With the potential of Blockchain technology more things can be done to help to the circular economy. It can help the circular economy in many ways such as helping to reduce transaction costs, enhancing performance and communication along the supply chain, ensuring human rights protection, enhancing healthcare patient confidentiality and welfare, and reducing carbon footprint [18]. Blockchain also can contribute to the circular economy by reducing hard copy usage, fuel usage, encourage energy renewal and substantially reduce waste in supply chains [3].

3 METHODS AND TOOLS

To develop our project and search for the necessary information, we have used tools that have been essential for the job. On the one hand, to search for information, we have browsed the internet using Google scholar³, where we have searched for articles, reports and papers on the subject we wanted to study and work on. Google Scholar has been practical for us, since apart from obtaining information, it allows us to copy links or cite the page directly in the overleaf, to facilitate citations in the texts. All the important documents that we have found, whether they are research papers, news, books, etc., we have been saving in the Mendeley⁴, a site that our teacher Fernando Vilariño recommended to us, where all the documents can be stored in an ordered manner, and can be consulted easily at any time.

For the search for information and to find answers to questions about some topics that we did not know how to solve, a book in particular has been very good for us, "Introducción al Blockchain y las criptomonedas en 100 preguntas" by Ismael Santiago Moreno [24]. This book talks about all aspects regarding Blockchain. It is a very useful book since it is arranged by chapters, where there are specific questions about each topic which are explained in a few pages.

In terms of methods, we used design thinking to identify problems and propose solutions that are relevant and solve the real needs of our users.⁵ We also used the Canvas business model (fig.9) and the SWOT model (fig.10) to get a better understanding of the environment and market around us.

Finally, we arranged an interview through video call with Ignasi Bonet Pocino, a social linker who connects challenges with solutions, supply with demand, in a

³google Scholar: <https://scholar.google.es/>

⁴Mendeley: <https://www.mendeley.com/>

⁵More information on design thinking: <https://www.designthinking.es/inicio/index.php>

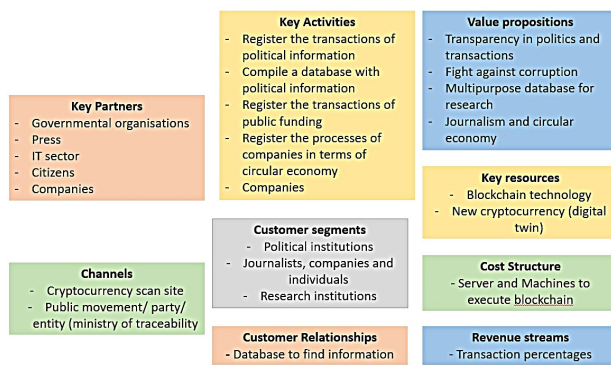


Fig. 9: Canvas business model of our project. Source: Own creation

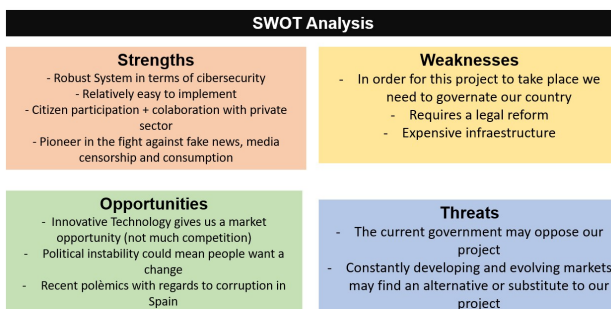


Fig. 10: SWOT analysis of our project. Source: Own creation

creative and innovative way. His interest is based on democratizing innovation and mainstreaming knowledge and facilitating the transfer of needs between citizens and public administrations. With Ignasi Bonet Pozino we had a talk where he explained to us what he did and in what aspects he used the Blockchain, and what benefits could derive from using this particular technology. Apart from answering the doubts we had, he helped us and gave us his point of view of our work, giving us some ideas on how to approach it and what issues to focus on according to its importance.

4 PROPOSAL

Our proposal, which we will further explained in the following section, can be broken down in 3 parts: the blockchain network in itself along with the token, which constitute the most important part and are the main focus of this paper, the database where political information can be found and basic notions about its design, and the political implications of our project, which will not be fully detailed in this paper since we would still need to clarify and structure our plans with experts before putting anything into practice. In order to solve the problems of fake news and subjectivity in real life events and in the other hand, the difficulty of tracking of all the public money, we propose a solution, a blockchain that mirrors all the public money and reflects every money transaction and where it goes (ministry, or any entity, or even to a politician) so everyone can easily track what is really happening with the money of the citizens, this can mitigate our traditional ways of

corruption in politics. On the other hand a blockchain is a database that stores info in a non linear way. Traditional databases store all the data in columns, so when a new data unit is added, it is added to the last place, but in a blockchain, it is different, it is stored in blocks.

Using this technology we can assure that a whole network will be protecting the integrity of the data. The more nodes, the more secure it is. Also anyone that belongs to the country can help, even any business can help and provide some computers to help the network.

Another strong point of this solution is that anyone can access the database and check any block or transaction that has been just verified or completed in the past.

So without further ado, let's analyze the different components of our proposal:

4.1 The design of the network

The blockchain privacy will be hybrid, it is a combination of a public and private politics in terms of participation in the blockchain. First of all, at terms of who can participate: only citizens that are identified in a country (in this case, Spain), as a public entity, we believe in the principles of decentralization, but, everyone that is helping should be identified - "someone that would never do something bad for the system would not mind to reveal its identity". Otherwise, someone with a hidden identity is more likely to commit fraud.

And also, to work and help in this network, you need to belong to the country, it makes no sense to allow a foreign government or outsider entity to get involved in the process, it's a government business, and other governments may try to weaken the blockchain.

Summarizing, only citizens identified inside the country can help the system, but anyone else can check what is going inside the blockchain.

4.1.1 Protocols and proof of work

We have seen in previous chapters which 2 protocols exist: Proof of work (brute force) and proof of stake(validations).

The protocol of this blockchain will be proof of work, with a block mining rate of 10 minutes. In PoS, the miners are validators, and it requires a much less superior computing capacity and so the electricity consumption gets reduced, making it greener for the planet. We do not need an army of computers spending x10 times the electricity we would need to maintain the system. PoS is the solution to reduce the electricity consumption, and also reducing the computers number of the system, it is true that this protocol is not as safe as the proof of work, but we make sure that if someone does any malicious action to the blockchain then he/she will be identified and punished.

4.1.2 Participation

The main idea is that anyone inside the country can help, from other institutions to particular citizens and medium/big businesses.

Starting from the enterprises of the country, a business that helps the transparency of the public money tracking

will have a good opinion or view from their consumers. The public opinion and how the society sees a business is a very important thing to pay attention, so any corporation that is willing to improve their social opinion can help on this blockchain offering a part of the computing capacity of their computers, as we said, PoS (Proof of Stake) is a protocol that does not require a farm of computers, just some validating transactions and verifying the blockchain.

Apart from corporations, any single citizen willing to help with their personal computers to the blockchain will be accepted. The idea is that anyone can participate.

4.1.3 Corporations in Blockchain participation

An agreement between the administration and the 50 biggest companies of the country can be made, and they may be obligated to participate in the process of verifying, for the benefit of the whole population. A big company in a country can easily make a lot of millions per year, so it will not be a challenge for them to offer a few computers to serve the blockchain, at the end of all, they are the ones that pay the most taxes and the ones powering the economy. It makes sense that they must contribute to public money tracking, because at the end of all, they are the ones that contribute the most to this public money. And its their duty that the money from taxes goes to the good hands.

4.1.4 System rewards

As any economical transaction, A works for B, and B rewards A for the task done. In this scheme, A is the group of citizens, corporations and businesses and B is the state. The state will reward the verifactor that completed the block, this is chosen randomly, as PoS is a random reward protocol, the more CPU someone gives, the more chances exist of “mining” the block. This currency will be euros(€) or the legal currency of that state. And it will be also reflected inside the blockchain, because a portion of the maintenance of the Blockchain ministry will be to sustain the reward system.

As we said before, in Proof of Stake, the miners are verifiers, and to be trusted by the system, they have to stake an amount of money to participate in the network and start verifying transactions. But, in our blockchain, the more resources you want to offer, the higher amount of stake you will offer the system. This is because a single person that wants to offer 2 computers to the system does not need to pay the same as a big corporation that offers 48 computers. It has to be reachable for anyone. And the more computers, the more damage you can do to a network, so it makes sense to offer a higher amount of money to lose in case you try to vulnerable the system.

4.1.5 Machines design and laws

The machines that help the system will be categorized in two types: Governmental and Private. This difference is to split and identify which nodes are from the state and which not, the ones from the state will have to accomplish all the rules required to start verifying, while in the private category, they do not need to accomplish the rules/laws.

The set of rules for the administrative nodes are the ones similar to a DPC (Data protection Center). They need to have a good response against: Electric cuts, fire, inundations, heating due bad performance of air vents, particles that damage the electronics and air vents, humidity in excess that can provoke water condensation, non authorized access (hackers and breaches in the system), and physically well protected and located inside an administrative building (government, ministry, police station, firefighter station, school, hospital...).

The governmental nodes need to have a non interrupted electric system to not have a computer shutdown, a video camera surveillance system that records everything inside the DPC and their workers inside, also we need sensors in the doors and devices inside, a fire detection system, climate sensors that measure humidity, temperature, smoke detector.. And needs to be located inside an administrative building, why? Simple, because of protection, someone that wants to get inside the DPC, will need to get inside a police station or hospital and break all the security levels, and then break inside the room, if it wasn't like this, then the computers would be very exposed and vulnerable to physical attacks, and all of the will have to copy every block in a hard disk (so in case of attack we have a lot of copies of the blockchain).

Apart from that, all the computers will have to accomplish a specific set of rules, like being imported from industries that belong to the country (to protect the economy of the country) . Also they cannot surpass a pollution level in terms of which pieces they are made from (plastics, screens, wires...).

4.1.6 Why the State centralizes the decentralized

Our system uses Blockchain technology, an anonymous decentralized technology and turns it into a centralized and user identifiable technology. The question is: Why?

First of all, it is a matter of security, innovation can be dangerous and even more so if it is financed by public money, for the administration a safe path is and will always be better than an uncertain path, and so, the approach and integration of this new technology. They have all the public money and they need a 99% success rate, they cannot fail or invest some money that will never return.

So they have to grant security to the citizens and this is why every node has to be identified, as we said before “someone without malicious intentions would mind to not be identified, while someone that is doing the good, will not”. It is a matter of public safety. In case of an attack, we will know who it was.

About centralizing the decentralized: The blockchain is hybrid, not private but not public, and it's under the care of the Blockchain ministry. They allow or kick users in the network and they manage the identification of every node. It cannot be a public blockchain because otherwise we would have “too many breaches to cover in this ship”. But hybrid is the best choice, but the only difference between this and a private one is that the book of transactions is open to anyone.

To follow a technology does not mean to follow their ideals or foundations, we are just picking their best and adapting it into our problem to create a solution.

4.2 Political implications

The ambitions of this project are high, meaning in order to put all our ideas into practice we would need to completely reform the system as we know it today. A legal reform without precedents which would put Spain in track to become one of the pioneer countries in terms of transparency, sustainability, citizen participation and technological innovation would be needed for our project to be possible.

For this reason, seeing the way in which Spain works, for this idea to come to reality we would need to reach the power, preferably with absolute majority and the support of a new political party who could help us reform the nation and put an end to the status quo that has been dominating the country since the transition to democracy.

The implications of this are essentially that, in need of a new political party, we would have to create one or lead a revolution to reach the power. For practical purposes we think the democratic option is the better of the two. Since none of us is an expert in the matter, we would first need to find associates with high knowledge on the field who could join us in our way to reforming the country and investors to support our campaigns.

Supposing we finally made it to the government, and again acknowledging that we are not experts in how to lead a country, we would still need to have at least a vague idea of how we would ideally like to approach our mandate.

The first step is deciding how the government itself would be structured. We fundamentally have two goals which are making politics and the public economy more transparent, and governing the country through experts who have extensive experience in their fields of work and can make better decisions than us, who are clearly not competent enough in all aspects involved in governing a country to make it prosper in the way we want it to.

These two goals quickly become problems when we look at the current political system in Spain, where transparency has always been questionable, as proven by the many corruption trials that have taken place in recent years which expose the possibility of politicians committing irregularities with public money. We also see an issue in the profiles of the people who end up in the parliament, which are in their vast majority career politicians, and those who manage to form government generally end up relying on experts to give them counseling as their advisors, who usually don't have much authority in terms of decision making and executive power.

As with all problems, if we want to end them we need solutions, so here is our proposal to solve these two issues we have found.

4.2.1 The new political system

In order to be able to effectively implement a Blockchain at a national level while implicating the public administration, private companies and particular users we need to ensure a strong government which will put the interests of the country as its first priority, rather than the interests of the political parties. This leads us directly to questioning if the current political system is aligned with our goal of revolutionizing the way a country can be governed, focusing on practical and pragmatic solutions with a notable

impact on the well being of the population and not making decisions based on the consensus of a political party which marginalizes the differing opinions of the opposition.

In this case, we find democracy to be a flawed system due to a series of weaknesses which would be exposed when trying to implement such measures.

The main flaws of democracy are the overcharge of the system and the polyarchy. The decision making in a democratic system is slowed down by the democratic process. The wide range of approaches the different parties can propose to tackle the same issue means the voting procedures can become stuck and not be resolved if an agreement isn't reached on how to solve a problem. This also means that ultimately the topics which concern the majority of the voters are always prioritized over those which concern only a minority of them. If we extrapolate this to political parties, we get the results seen in many countries where the governing parties do as they please and often ignore the ideas of those in the opposition. In some cases, this means that ideas that could benefit the population are overlooked in favor of the party's interests. After all, we can affirm that the political capital is concentrated in a few circles of power with almost full capacity to make decisions. This poliarchy concentrated in these circles of power also creates a certain competition between them, leading to a conflict between those who supposedly have to solve problems, which once again slows or blocks the process of democracy.

In conclusion, if we add all these factors up, we end up getting a system which is theoretically based on giving power to the people where the majority always rules over the minorities, and it ends up creating a chaotic mess where conflicts end up being resolved in an almost authoritarian manner where the few who accumulate the most political influence are those making all the decisions.

This is reflected in the annual State of Global Democracy index published by EIU.com (afegir enllaç), where Spain has fallen from being a full democracy to a flawed democracy, finding itself on a descending trend since 2019 and falling out of the full democracy range for the first time ever since the index was created. In other words, this is a signal of the country being a victim of these flaws of democracy and not being able to correctly adapt and overcome the multiple issues rising in Spanish politics in recent years.

For these reasons we propose adopting a new system: Technocracy.

A technocracy is a form of government which is based on the idea that those who make decisions are experts in their areas of responsibility, so they are not elected through a voting process but rather selected based on their expertise, especially in technical or scientific fields.

During history there have been some states which attempted to flirt with the technocratic system, but none of them went all in and replaced their current political structure for a technocracy. For example, the Chinese government is filled with technocrats with science or engineering degrees, but the political system is far from a technocracy. Another example would be Italy, where as of May 2022 the government is led by technocrat Mario Draghi, who was appointed in 2021 by former president Sergio Mattarella, but in this case only 8 of the 23 ministers

are technocrats who are not affiliated to any of the current political parties in the country.

In our case, we contemplate the idea of fully committing to this new system, but it is clear that a transition like this would create some discrepancies between the population, and especially turn the career politicians against our movement. To avoid this, we propose a hybrid system, close to the Italian model but with less representation of career politicians. In this case the Head of State would remain His Majesty King Felipe VI, while the president would be elected through a democratic process just like it is now, but with the exception that only 25% of the ministers could be career politicians, and the remaining 75% would have to be selected from a pool of candidates based on expertise by the new Political, Economic and Social Traceability Ministry, which would be composed entirely of technocrats (for more information see point 5.2).

In this new system the senate would be eliminated and all the executive and legislative powers would be competency of the new Congress formed by the technocrats and the party representatives in the government, with the Supreme Court and the Political, Economic and Social Traceability Ministry being in charge of the judicial chamber to evaluate laws concerning their respective competencies.

4.3 A reliable source of information

As discussed in point 2, it has become clear over the years that manipulating information both in terms of what is presented and what is hidden and how it is presented plays a crucial role in how the general public perceives that information.

All traditional media has the problem of offering data that is not fully accurate due to the bias that exists when writing an article or presenting the news, whether it be the way in which the titular is written to fish for views, the political views of the author making a theoretically objective article slightly biased as a result, or the unreliability of opinionated articles that doesn't allow the readers to contextualize or compare what's being told to them unless they make an extra effort to contrast the article, which is quite uncommon.

For these reasons we have the strong belief that intoxicating data with the opinions of a journalist, who is often times not an expert in the matter, or if they are they will probably be biased, is a source of information that can be exploited by governments and circles of power for their own benefit, and therefore we need to ensure that there is a way for those looking to find the most objective data possible to do so with ease and without having to go through an intensive process of verifying the reliability of their sources, contrasting information and reading from multiple sources just to find the slightest piece of data.

In addition, we also want to take this opportunity to concentrate all sorts of data provided by a large number of organisations in one place, offering a service where you can quickly find any relevant contrasted information about any political or social topic with as many entries as possible, all in one place.

4.3.1 What will you find in our database?

The main goal of this project in terms of offering transparency and eliminating bias from news is to offer as much objective data as possible, gathered by experts in the matter who can comment as a group with different opinions to guarantee that a single author cannot show their bias.

For this reason we think the most convenient and accurate way of organising the information in our database is by identifying three main types of data.

The first type will be numerical data. In this group we can find all types of objective data that has been gathered from elections, referendums, voting processes in a congress, senate, government,... Indexes published periodically by organisations such as the UN, EU, NGOs and such will also be considered numerical data as they offer a classification of elements through a system designed by the creator of that index.

An example of this kind of data would be the exact results of the 2020 presidential elections in the USA, with the ability of sorting the results at a national level, by state, by county,... Another example can be the results of the voting phase to pass a certain law in a democratic country, or the press freedom index published each year by Reporters sans frontières.

The second type of data will be reports. These mainly consist of the detailed explanation of the indexes commented above, as in the full document that breaks down the process of creation of the index and the analysis of its results. Other reports have their focus on extensively analysing a certain topic and giving the most accurate results possible. We ensure that these reports aren't biased because they offer objective data found during the elaboration of the index, and the sources are widely considered to be references in their respective sectors worldwide.

Examples can be the UNODC world drug report published yearly by the United Nations Office on Drugs and Crime which analyses the global drug market, or the Freedom in the World report that's published yearly by Freedom House, assessing the condition of political rights and civil liberties around the world.

The third type of data are research articles published by organisations such as national institutes, colleges or experts which focus on very specific political and social matters and offer intensive research through quantitative or qualitative methods that comply with the standards of the scientific method used in social sciences.

These articles are nearly infinite, and their reliability is also the hardest to determine out of all the categories we have identified, so regulating these will be a much bigger issue than with the reports and data published by official institutions that's almost certainly going to be objective.

4.3.2 Verifying the information

While we could consider the possibility of including only information from sources widely regarded as reliable like the UN, we don't want to limit the range of knowledge in our database, so we need a way to verify the reliability of our sources that may not be well known, as well as sources such as research articles or papers that may come from an

individual or a group and aren't tied to any organisation or institution.

To achieve this we will need an organisation similar to those agencies currently in charge of verifying news articles as "non-fake news". For our purpose we will need an organisation that can take on a much larger quantity of data, check the method in which the information has been obtained

4.4 The token

As we know the cryptocurrency is the engine of the blockchain technology. In our case, it is mandatory to have a system to control public transactions and make them transparent. We have planned to create a digital twin of the Euro through stable currency as a mirror of public transactions. However, the aim of stablecoins is to convert the physical money to digital money and it is not what we want because our objective is not to eliminate money as we know it today, but to replicate the movements in a blockchain to make it visual (relation with citizen science). The advantage of the stablecoin is that the value remains constant and it's perfect for our case because it will replicate the value of the euro. In addition is important to distinct the two types of stablecoins in order to choose the correct one for our project:

- Retail stablecoins: assets available to a wide range of users.
- Wholesale stablecoins: assets available to a limited circle of users

Keeping in mind that our system will be divided in three different groups of people: Financial traceability ministry (controls and replicate the transactions of public money using the stablecoin), public administrations and citizens (the users of the platform), the good option would be Retail stablecoin due to the huge scope of population.

5 APPLICATION

5.1 Graphic Design

As we can see in this picture, this would be a graphical concept of the node type distribution. In the legend there are different nodes. The strengths of this distribution is that against a specific physical attack to the government installations, the citizens nodes and companies nodes would still support the network integrity (making it more vulnerable, but still safe). So even a coordinated strike attacking the physical administrative buildings of the country, this network will still work. "A blockchain for the country, sustained by the country" This would be the definition of this concept, that the whole society would participate in it. Minor attacks to some companies can also affect the network but their changes will be minimal. Citizens nodes attack too. We are contemplating every scenario, even an attack of a missile can be possible, so the more nodes the better and this is why the whole society participation is a must. This method and proposal of this network example could be implemented in the future for

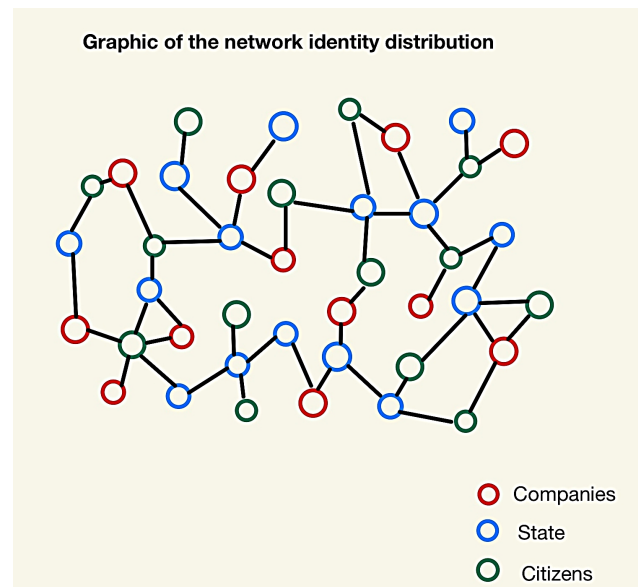


Fig. 11: Graphic of the network identity distribution. Source: Own creation

data integrity or any other public data, centralizing the info can be very risky, once that place or vault is sabotaged, all the content is exposed. But decentralizing it in a p2p network can be the best option to maintain the network integrity.

5.2 Mining/Verifying Rooms Graphic Design

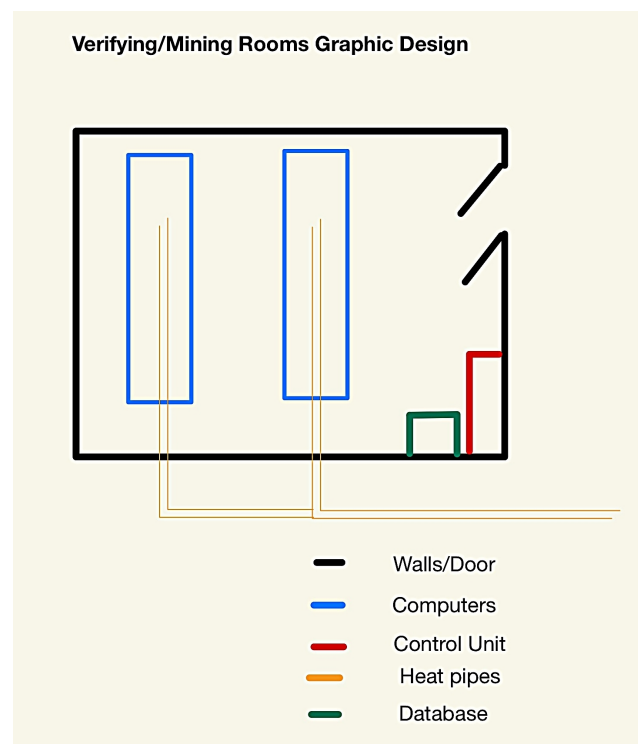


Fig. 12: The mining rooms Source: Own creation

As It's said before, in administrative buildings the miners/verifiers have to accomplish some conditions mentioned before. This graphic represents a room where our computers would be located. We can see two rows

of computers, each row will have 12 computers with their multiple integrated graphic cards to do a better performance at verifying. A Control Unit will be located inside the room, where we can see the status of each of the computers, and if needed, the computer engineer will assist the devices that need help. Inside the same room, there will be a database in which an algorithm will add every block of the blockchain to a linear database Blockchain: Distributed database organized in blocks, nothing is linear. Traditional Database: Everything is linear, every new entry is added after the last addition. Our objective is to translate this blockchain database into a linear/traditional database that for the user will be easier to read info instead of searching every block transaction. This database is also a backup copy in case of a massive loss of the network or conflict, so we can restore it and not lose any data. This is the backup plan if everything fails. Every week this database will be updated with the current blocks of the last week, so we can keep safe the new content and we have time to verify the news of the current week before they can be confirmed. Everyone will be able to check the blocks in the blockchain or in the database, depending on the user search skills in these two technologies: Block tracking or database search. We can see a heat conductor that will absorb all the hot air produced by the computers to heat the administrative building in which our node is located

5.3 Regulating the network

It will be important to create an institution with the aim of controlling our blockchain network. This organism will be called Political, Economic and Social Traceability Ministry. This new Ministry in Spanish government will focus on the traceability and transparency of the transactions made with public funds (Financial traceability), the regulation of the environmental impact of the companies participating in the Blockchain network (circular economy) and the verification of sources that can insert content in the blockchain's political database (Transparent information).

The entity will be in the top of our organization chart, meaning it will be legally able to impose sanctions and filter any content that goes into the database, as well as decide who can and cannot contribute to the blockchain by adding computational capacity.

The main goal of the institution in terms of financial traceability is to replicate the public transfers with our stablecoin. Furthermore, it will be in charge of making the records of those transactions accessible for the general public to trace what has been done with their public funds.

In terms of circular economy, the Ministry will be in charge of enacting a new law concerning sustainability in data protection centers and blockchain mines, while ensuring that corporations, individuals and public entities which contribute to the Blockchain are complying with these new regulations.

As far as transparent information goes, the role of this new Ministry will be validating and verifying sources of information following a method which is going to be developed by experts which rates information and classifies it based on its reliability, and validates sources based on the average reliability of its most important contributions to transparent information. This department will also enable

complaints and reports on information that may have not been filtered correctly on a first instance so that readers who can support their report claim with evidence can help us keep the database clean of opinionated articles, misinformation and erratic or factually incorrect data. The distribution or the hierarchical structure of the new ministry could be represented like this:

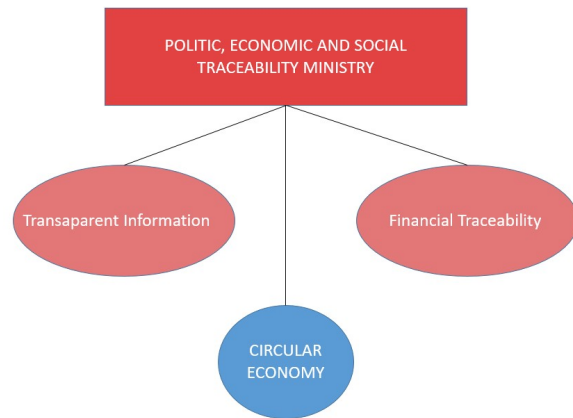


Fig. 13: The Politic, economic and social traceability ministry

Source: Own creation

5.3.1 Objective news insertion inside the Blockchain

The department of transparent information is in charge to verify all the information and insert it in the blockchain, this will be in plain text and the only digital files that can be attached in the blockchain will be photos, for now. This news will be able to be searched inside the database once they are verified after a week. It is hard to say if something is right or not the same day it happens, the more testimonies exist the more trustable that information is. The transparent information department will be in charge of confirming the news of that week so they can be permanently added inside the blockchain. "Something is right is the majority says it is right, same happens in a blockchain" News cannot be superior to 6000 characters and not contain files bigger than 10 MB, every news can contain up to 8 files. In every block there can only be 2 news inserted:

- 2 news every 10 minutes (10 minutes is what it takes to mine a block)
- 12 news per hour; 288 news a day; 2016 news a week

We think this is enough, later on, the network can get updates to increase or improve this, but for starting this is enough.

5.4 Creating an engaging interface for our application

When developing a user interface (UI) it is important to anticipate the needs of the users to ensure an easy, intuitive and comfortable navigation.

First of all, we need to consider the elements present in our interface and attempt to implement features with which

the general public is already familiar. As suggested by the US General Services Administration ⁶, the best practices when creating a UI are the following:

- Keeping it simple
- Creating consistency within the UI
- Using colors and textures strategically,
- Creating a hierarchy using typography
- Considering spatial relationships between elements
- Communicating the status of the UI
- Adding defaults

When thinking of keeping it simple, we immediately came to the conclusion that the simplest, most familiar way of approaching our UI is to make it similar to those browsers which people use the most. Taking the Google browser as an example, our database could have a main search bar and present the results sorted in some sort of hierarchy. In our case, we believe the most convenient ways of sorting results would be by reliability (using the rating developed by the experts in our Ministry explained in point 5.2), by date (sorting by the newest or oldest elements chronologically) and by relevance (the most consulted elements being on top). The default way of sorting results would be reliability as it is the most aligned with our aim of bringing accessible, reliable information to the public.

When creating consistency within our browser we think the best way is to use common elements in the UI and keeping the layout of the browser uniform. This leads us to the typography, where we should use one font exclusively to facilitate reading, and create a hierarchy using letter size, bold letters and italics depending on the importance of the items in said hierarchy we are trying to create. Keeping consistency in the colors we use is also important, not changing the background of the browser constantly, having everything in the same hierarchy level displayed in the same color and changing the color of the entries that have previously been consulted to facilitate the process of research.

The spatial relations between elements are also key so that users can distinguish different items in the results tab, so we must avoid the compression of elements in a reduced space or too much spacing which would make it uncomfortable to scroll down constantly just to see a couple more results.

In addition to the more visual side of the UI, we also need to add shortcuts to improve the quality of life of the users trying to access certain features, like changing between browser pages quickly or going back and forth between consulted pages.

Last but not least, we need to be transparent in terms of the status of our UI. If anything is being changed or isn't working properly users will be notified, there will be some sort of help or assistance tab for those who are trying to use in-depth features such as advanced filtering of results or for those getting started with the use of the system, as well as explaining basic concepts of what one can find in

our database. We also believe adding a feature for people to suggest improvements to our UI through polls or comments is important for the health and quality of life of the browser.

6 DISCUSSION

In the following section we will analyze our project from 4 different perspectives:

- Citizen engagement
- Citizen science
- Circular economy
- The quadruple helix

In each of the subsections we will first introduce the concept we will be analyzing and then providing a point of view from that perspective of our project.

6.1 Citizen Engagement

Our goal as a new movement advocating for transparency and fighting against misinformation is to increase the citizen engagement since we consider it to be a crucial part of reaching as many people as possible with our system. It is key that we have a comfortable and user-friendly interface, as well as different functionalities that may be implemented during the life of our project, which will be modified according to quantitative analysis involving the citizens who want to obtain a benefit through our service, because in the end they are our target customers. For this reason, we keep in mind the five guiding principles: Keep it simple, Make it visible, Make it personal, Make it practical and Make it playful.

We will start with "making it personal". Our purpose is to reach citizens in a personal way, allowing us to know how our project will affect them and thus being able to identify how they can find an advantage and benefit from it.

The term "make it visible" aims to deliver information to citizens in a simple, comfortable and easy to understand way.

The term "Make it practical" is very important for us, because in order for people to trust our project, and any other project for that matter, one has to be able to demonstrate a practical application so that people can really see how it can benefit them and how it could potentially change their daily life for the better, so that they engage with us in further developments and improvements for the system. In this case, we want to "Make it playful" because we believe it helps us achieve a result that is inspiring and fun for citizens, so that they feel comfortable with us and can contribute to new solutions and invites them to share our project with their circles.

Last but not least, "Make it simple", explain in a clear and simple way what the project itself consists of and how we want to help people with it, so ultimately what benefits they can gain from it.

So after reviewing the characteristics of citizen engagement we can conclude that our project will indeed meet these requisites. We will make it personal because we will offer citizens the power of instantly checking how

⁶US General Services Administration website: <https://www.usability.gov/index.html>

public money is being used (which is a direct result of the taxes they pay, so they should have the right of monitoring all transactions), and knowing what purposes politicians use this money for, in order to ensure that it is used wisely and in a way that is aligned with the general public opinion. We will make it visible by implementing our website/application where our political database and public fund tracing are shown, where citizens can execute all the searches in a simple and visible way, so that it is not a tedious job to be informed. We will make it practical, simple and playful through this website/app, which will have a concise and intuitive UI, and allow people to engage on mechanisms to suggest improvements to the system.

6.2 Citizen Science

For the creation of new projects it is necessary to innovate, generate ideas that have value so that sustainable results arise which have a common benefit, both for the planet and for people, where the quality of life can be improved. Science has helped a lot with this, but it has also evolved a lot. Nowadays, science is much more collaborative, it allows everyone to participate in a new project, contribute with new ideas and show new points of view that had not yet been contemplated. This gives the opportunity for these projects to be more elaborated, and to have a much greater consistency, since the more collaboration there is in a project, the more points of support there will be in its basis and more convincing and compact conclusions can be drawn.

Citizen science work is carried out by members of the general public, it describes a wide range of activities, in which everyone can participate in a scientific project. This means that the solutions, ideas and motivations of the project are more varied than if they were made in a studio. In Blockchain, citizen participation is crucial for its operation and development. Especially the collaboration of companies that use our system and the society that surrounds us.

For this reason we have the intention of implementing feedback systems as described in the UI development of our database (5.3), and we will also allow particulars to contribute to the computational power of our blockchain while not having to comply with the same strict regulations that will be imposed on private companies.

6.3 Circular Economy

At first glance it may seem that our project has nothing to do with circular economy since it consists mostly of virtual components, but in order to make all of this work, as seen in points 4 and 5, the implementation of a huge infrastructure is required.

For a blockchain to be operative, especially one of the magnitude of this one, a large amount of high-performance computers using the latest technology is required. When these computers are no longer able to work at their full capacity, they have to be replaced by more modern options, which obviously leads to a problem; what do we do with the old computers we cannot use anymore?

As much as these machines become obsolete for mining our blockchain, they are still undoubtedly incredibly powerful computers which still have a long life ahead of them. For this reason we believe it would be a waste just to throw them away or recycle them right away, and instead we will donate all computers which have been used in the blockchain and belong to the government to entities which can give them a second life and extend their life for many years. Examples of this can be elderly care centers, town halls in need of new computers, public schools or public libraries.

In addition to this, we also urge private companies and individuals to do the same, or at the very least sell their obsolete machines to others who can give them use rather than throwing them away. Campaigns to promote this practice will be conducted by the circular economy department of the Political, Economic and Social Traceability Ministry.

But even if we try to extend the life of these machines as much as possible, the issue of them becoming completely unusable some day still remains, so all the computers owned by the government that will be a part of the blockchain will be made entirely of recyclable materials and ready to be separated into different parts and materials which can be used for other purposes. This will also be suggested to private companies and individuals who contribute to the blockchain but not compulsory, at least in the near future.

Another issue with having huge security rooms full of high-tech computers working at full or at least high capacity is the heat that they emit. For this reason we will enact a law that forces all security rooms owned by the government to redirect the heat generated by the computers to nearby office buildings through the ventilation system.

Another measure we will adopt is using energy generated exclusively from renewable sources to power the computers, since they require large quantities of power and using fossil fuels to produce all of it would go against the core principles and values of this project. This measure will not be compulsory at first for private companies and individuals at first, but it is our goal to change this in the future to promote the use of green energy in our country.

6.3.1 Circular economy in our blockchain technology

So we want to make this as sustainable as possible, not only digitally, also physically, and we are speaking specifically of the materials. So a protocol of computer dismantling will have to be made, we cannot pretend to throw a computer when it is not useful anymore, the pieces can be reused or recycled, also, the computers that cannot mine anymore but still can work will be moved to other administrative buildings so the workers can work from that computer (remember that these computers have the latest technology).

An advantage of having this DPC inside a building is that we can use the heating produced from the computers to heat all the building instead of throwing the hot air outside, or make some kind of energy with this hot air.

Also, a percentage of the electricity that powers all the computers will have to come from a green energy source (solar panels or wind turbines for example).

Every part of this system has to be green:

- Computer physical pieces: Recycled pieces and dismantling protocols
- Energy Source: Green and renewable energies
- Algorithm: Proof of Stake
- Output: Hot air to heat the administrative buildings

6.4 The Quadruple Helix

Another important dimension to discuss in our project is the role of Companies, Administration, Academia and Citizens. These four branches are the key of our system because they are the engine of society.

Administration and companies will be regulated by our blockchain technology with the help of stablecoin in order to surveil the economy and politics.

On the other hand there is Academia. When we talk about Academia we are referring to research institutes, universities, highschools, schools... This branch is also very important because our database will be useful for researchers, teachers and students as a learning tool and as a source of verified information.

Finally, we have the citizens, who will represent the majority of the users of our system. As a citizen you can look for information and stay sure that it is verified and transparent.

Our system is divided in two types of permits: viewing permission and file editing and uploading permission. These four branches only have the viewing permission because they cannot edit or upload the information and the preferences of our blockchain technology. Nevertheless, there exists a small part of the administration, the Ministry of Financial, Economics and Politics Traceability that has the special permission because it's the organism that controls the well-doing and the information of the entire system.

In order to give transparency to the production process and use of the different resources of the companies, the new Ministry will carry out audits to verify all the processes carried out until reaching the final product. If the company pass the audits without discrepancies it will obtain a quality seal and will appear in the blockchain. In this way the companies that pass the audits will have added value and the people will trust in them. In other words, all companies that do not appear in the blockchain could indicate that they do not meet the requirements of the circular economy and people will be aware of it. However, it is not mandatory to pass these audits except in the case of IBEX-35 companies, who will have to pass it compulsorily because they represent an important part of the country in economic and environmental terms.

The auditors will be very strict and will focus in the entire process, from beginning to end. Child labor is prohibited, all workers must be insured with good working conditions and a salary according to the hours worked,... In addition, the different production processes must follow the protocols regulated by law and must pollute as little as possible, seeking new forms of production or the use of renewable energies.

The academy will be very important because it represents an important part of the consumers of our blockchain

technology. Our system is a perfect tool to use in many fields of study because it brings information related to circular economy, politics and news and public transactions. With all this information it is easier to write research documents and offer students real information without having been modified by different means.

7 CONCLUSIONS

In conclusion, the implementation of our blockchain technology system would be impossible without the proper reform of politics, economics and society. Given all these facts we think that the implementation of our idea would be complicated and complex due to the different interests of the whole population, and especially because of the established political class in Spain, which would undoubtedly impede the transition to the new transparent society we are trying to reach. In other words, we are talking about an Utopian idea of a transparent society. Also, the nature of the blockchain technology is transparency and decentralization, these are the opposite values of some democracies of today. It will take time for a state to implement this technology to solve a problem we have proposed. Even if we split the problem (public money tracking) and the solution (blockchain) we think no state would like to touch any of these topics. But in the near future change can be possible and one day this proposal of solution can be materialized. Even so, we think it is a solid proposal applicable to any government or administration of today and we have the right tools and the technology to do it. If this change is declined it is clearly because of a matter of interest (political, economical...) or because there is a better solution to the problem. Why do we have to use Blockchain technology for this problem? Basically, any citizen or company can validate what is happening inside the blockchain, and approving or declining transactions, so once a data is introduced on the blockchain, it cannot be modified again, for example a corrupt politician can delete some money movements in a device (computer, phone) in order to hide them, but in a blockchain it is not that easy, because every block is linked to the next, and so, you have to modify the next one. With this solution we prevent someone from deleting money movements, once they are on a blockchain, they will remain forever there. Although we are very satisfied with our proposal because we think it efficiently solves the problem. But a problem can have many solutions and this is ours.

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